PES2UG24CS044

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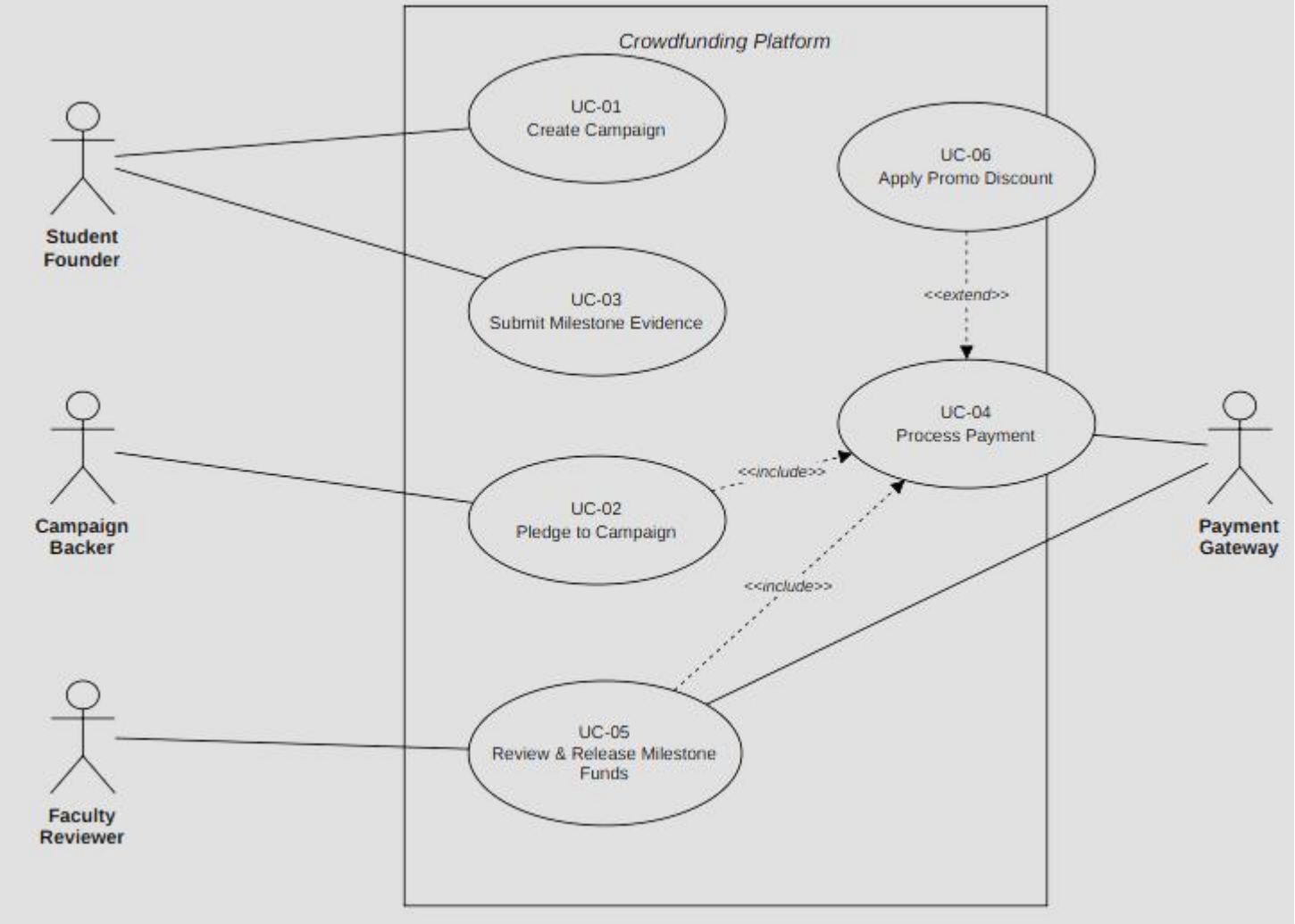
Problem Statement #40: Student Startup Crowdfunding Platform

REQUIREMENTS TABLE

Brief scenario: The system is used for student startup crowdfunding. Students can create campaigns, backers can pledge money and choose rewards, and funds are released only after the required milestone is checked by the faculty committee.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall collect backer pledges and release the money to the student venture in parts only after the required project milestone is verified.	High	Pass if the milestone is reviewed and approved before the tranche is released. Fail if money is released without approval.	This is the main control for releasing campaign funds.	Given in the problem statement.
FR-002	Functional	The system shall allow a Student Founder to create a campaign by entering the venture details, funding target, deadline and reward tiers.	High	Pass if a campaign is created after all required details are entered. Fail if the system accepts missing required details.	Campaigns have to be created before people can support them.	Required campaign details should be checked.
FR-003	Functional	The system shall allow a Campaign Backer to view active campaigns, choose a reward tier and enter a pledge amount.	High	Pass if the selected campaign, reward tier and pledge amount are saved together. Fail if any of these are missing.	This covers the main action performed by a backer.	Reward selection should be saved with the pledge.
FR-004	Functional	The system shall allow the Faculty Committee to review milestone evidence and approve or reject a request to release a tranche.	High	Pass if the committee's decision is saved and only an approved request can move to fund release. Fail if an unreviewed request is released.	Faculty review is needed before funds are released.	The decision should be recorded for checking later.
FR-005	Functional	The system shall notify the Student Founder and relevant Campaign Backers when a milestone is approved or rejected and when a tranche is released.	Medium	Pass if a notification is generated after each of these events. Fail if an event happens without the required notification.	Users should know when an important campaign update happens.	Notification can be through the platform.
NFR-001	Nonfunctional	The payment processing gateway must follow PCI-DSS requirements and use tokenized payment authorization.	High	Pass if payment testing shows tokenized authorization and no critical PCI-DSS compliance issue. Fail if raw payment details are stored.	Payment information needs to be handled securely.	Given in the problem statement.
NFR-002	Nonfunctional	The system shall have at least 99.5% monthly availability for campaign and pledge services, excluding planned maintenance.	Medium	Pass if monthly availability is 99.5% or more, excluding planned maintenance. Fail if it goes below 99.5%.	Users should be able to access campaigns and make pledges most of the time.	Planned maintenance should be recorded separately.

UML Use Case Diagram using draw.io



USE CASE FLOW SPECIFICATION

Primary Actor: Campaign Backer

Secondary Actor: Payment Gateway

Precondition

- The campaign is active and accepting pledges.
- The selected reward tier is available.
- The Campaign Backer is able to make a payment.

Postcondition

- The pledge is recorded for the selected campaign.
- The selected reward tier is linked to the pledge.
- The successful payment is recorded and the campaign funding amount is updated.

Main Success Scenario

1. The Campaign Backer opens the list of active campaigns.
2. The system shows the campaign details, including the funding target, current amount, deadline and available reward tiers.
3. The Campaign Backer selects a campaign and chooses a reward tier.
4. The system asks the backer to enter the amount they want to pledge.
5. The Campaign Backer enters the amount and confirms the pledge.
6. The system checks that the campaign is still active, the reward tier is available and the pledge amount is valid.
7. The system sends the payment request to the Payment Gateway.
8. The Payment Gateway authorizes the payment and sends a successful response.
9. The system saves the pledge, updates the campaign funding amount and records the selected reward tier.
10. The system shows a confirmation message to the Campaign Backer.

Alternate Flow

- At Step 8, the Payment Gateway declines the payment.
- The system shows a payment failed message and does not confirm the pledge.
- The Campaign Backer is given an option to try the payment again.
- If the payment succeeds on the next attempt, the flow continues from Step 9. If it fails again, the pledge is cancelled and the campaign funding amount is not changed.